

SH1017 Physics for Engineering Mathematics

Computer Lab 2: Diffraction and Interference

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Q1: Interference from Point Sources

Method: Using `interference.py`, we simulate two coherent point sources and observe how the interference pattern changes with the separation distance d and by adding a third source.

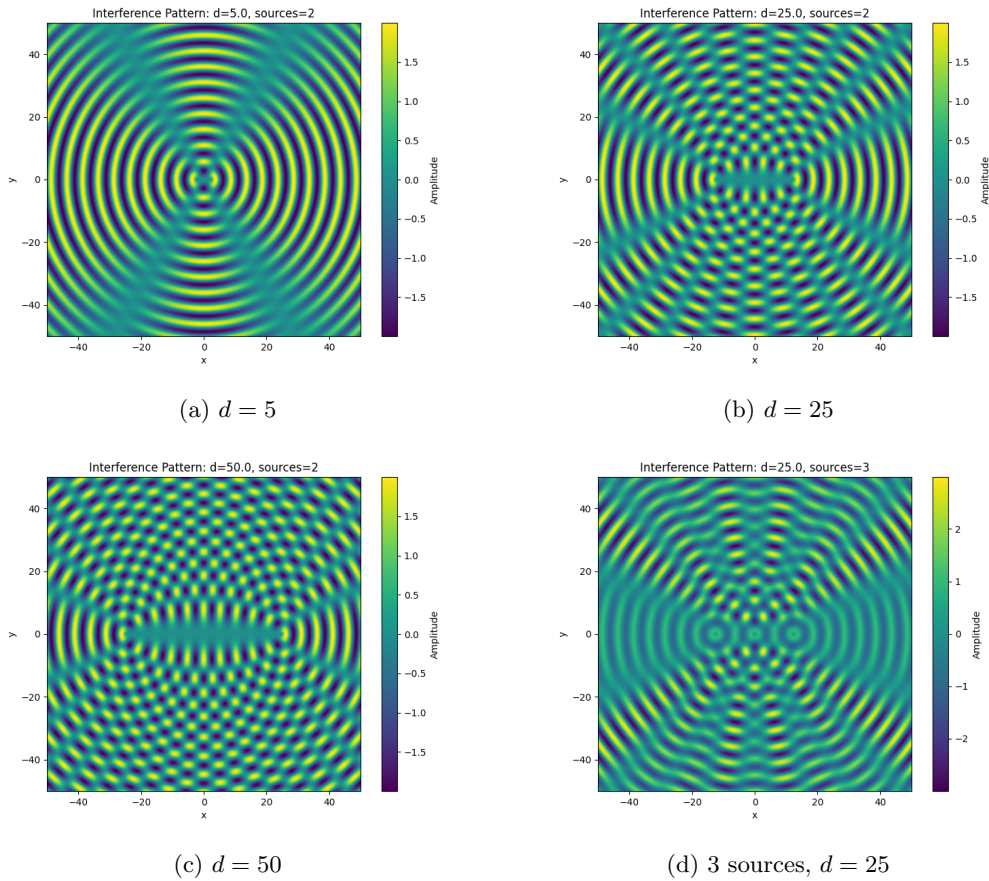
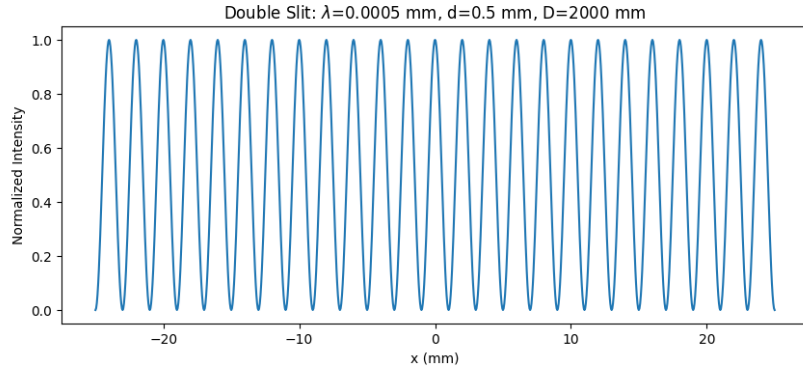


Figure 1: Interference patterns for different configurations ($\lambda = 5.0$ mm, 100×100 mm area).

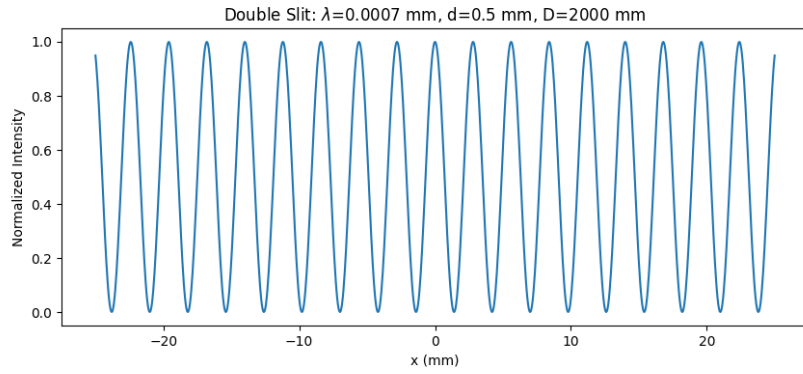
Results: As seen in Figure 1, increasing the distance d between the sources increases the number of interference fringes (maxima and minima) within the same area which corresponds to a smaller angular separation. Adding a third source between the two existing ones sharpens the primary maxima and introduces several more smaller maxima between them.

Q2: Double Slit Interference

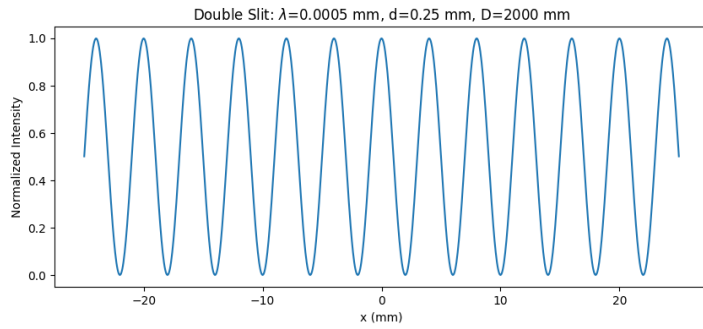
Method: Using the code from `doubleslit.py`, we simulate the interference pattern on a screen at distance D from two slits separated by d , using $\lambda = 500$ nm.



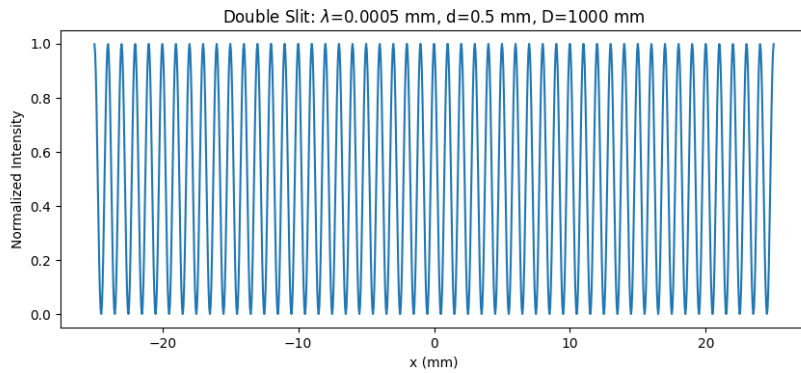
(a) Base: $\lambda = 0.0005$ mm, $d = 0.5$ mm, $D = 2000$ mm



(b) Larger λ : $\lambda = 0.0007$ mm, $d = 0.5$ mm, $D = 2000$ mm



(c) Smaller d : $\lambda = 0.0005$ mm, $d = 0.25$ mm, $D = 2000$ mm



(d) Smaller D : $\lambda = 0.0005$ mm, $d = 0.5$ mm, $D = 1000$ mm

Theory: The distance between maxima is given by $\Delta x = \lambda D / d$ where λ is the wavelength, D is the distance to the screen, and d is the distance between the slits.

Results: The results for various parameters are summarized below:

The numerical approximation of Δx is accurate to the theoretical value within 4 sig figs. And as

Case	λ (mm)	d (mm)	D (mm)	Theoretical Δx (mm)	Measured Δx (mm)
Base	0.0005	0.5	2000	2.0000	2.0000
Larger λ	0.0007	0.5	2000	2.8000	2.8000
Smaller d	0.0005	0.25	2000	4.0000	4.0000
Smaller D	0.0005	0.5	1000	1.0000	1.0000

Table 1: Comparison of theoretical and measured fringe spacing.

expected (see figure 2 and table 1) the distance between maxima increases linearly with both distance to the screen and wavelength and it decreases with distance between the slits.

Q3: Single Slit Model

Method: We modify `doubleslit.py` to approximate a single slit of width $w = 0.05$ mm by placing $N \in \{2, 3, 5, 10\}$ point sources along the slit, using $\lambda = 0.0005$ mm and $D = 2000$ mm. We investigate how the peak intensity and width depend on N and w .

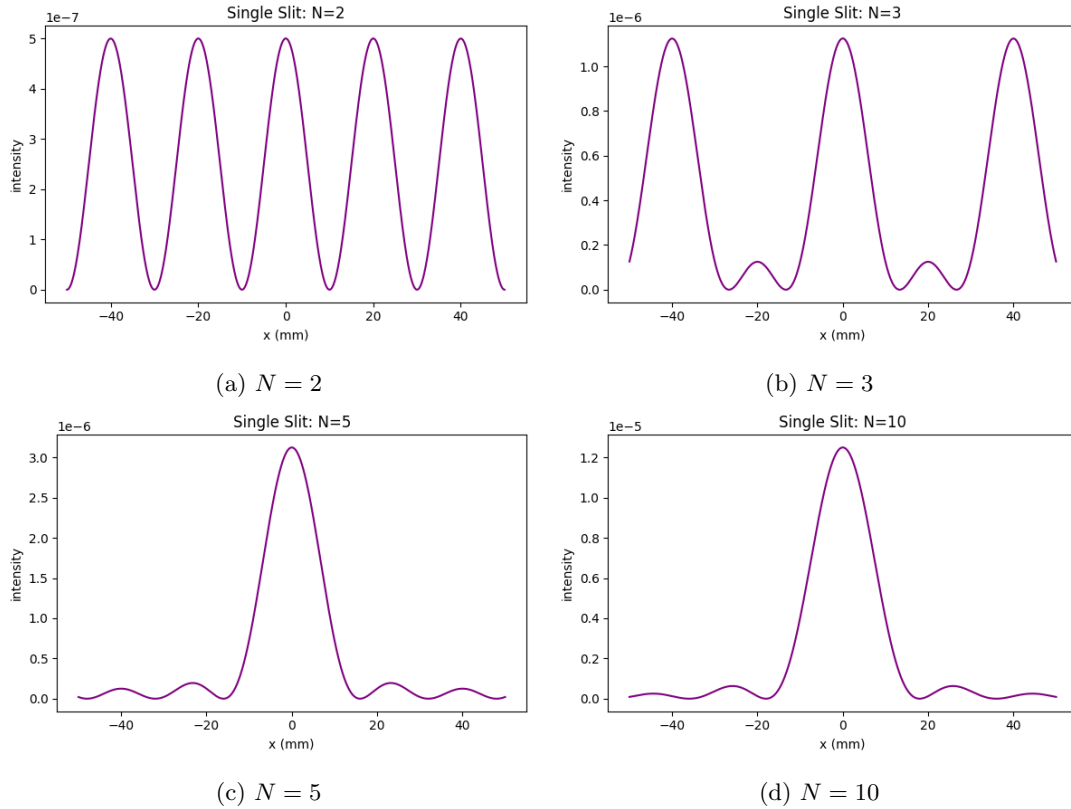


Figure 3: Intensity distribution for a single slit modeled with varying N sources ($w=0.05$ mm, $\lambda = 0.0005$ mm, $D = 2000$ mm).

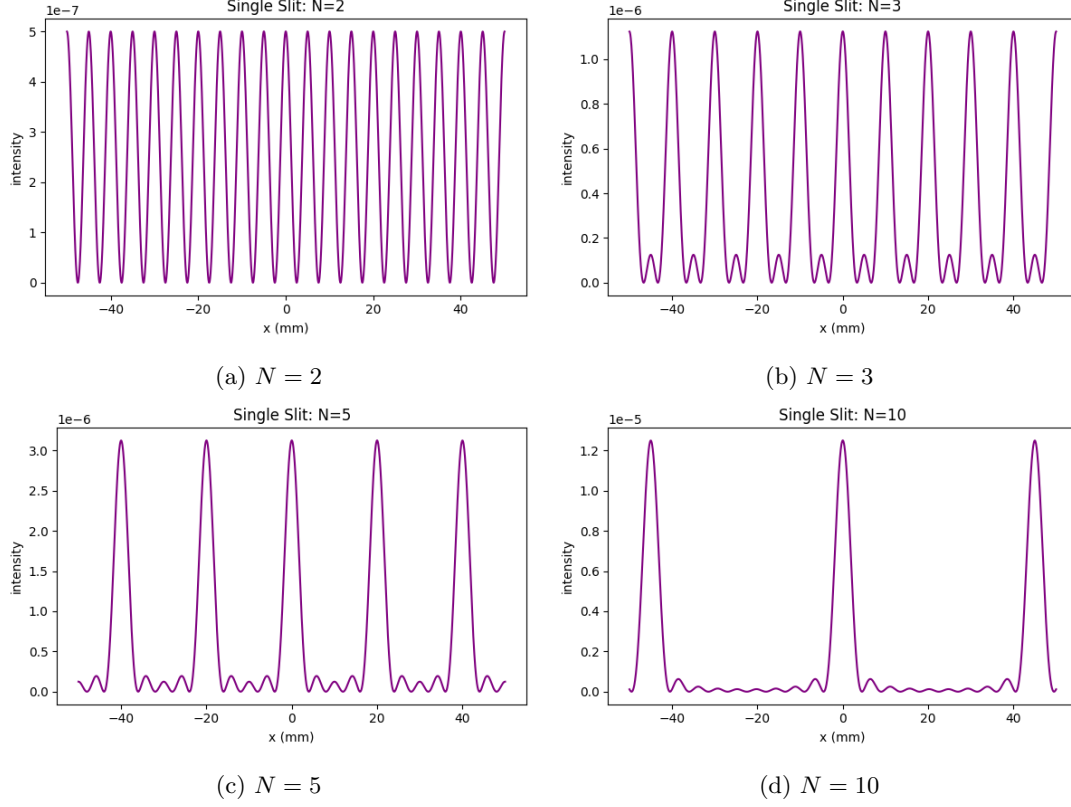
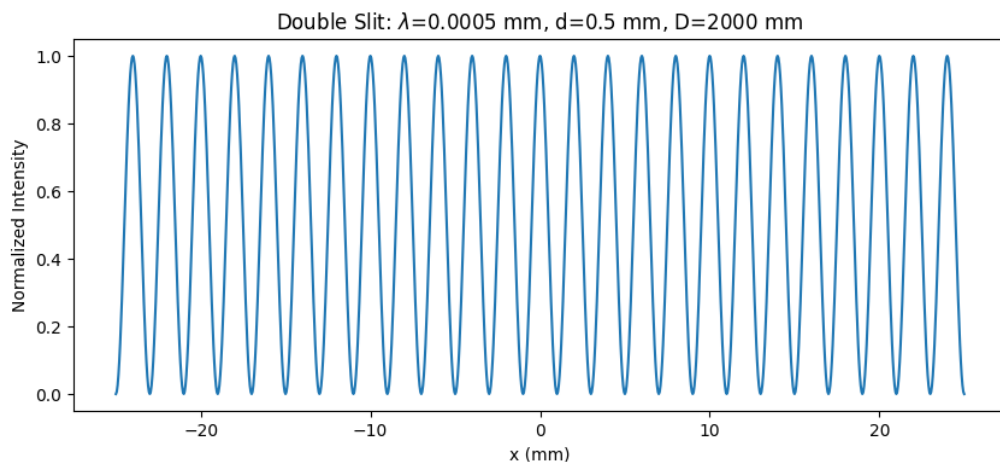


Figure 4: Intensity distribution for a single slit modeled with varying N sources ($w=0.2$ mm, $\lambda = 0.0005$ mm, $D = 2000$ mm).

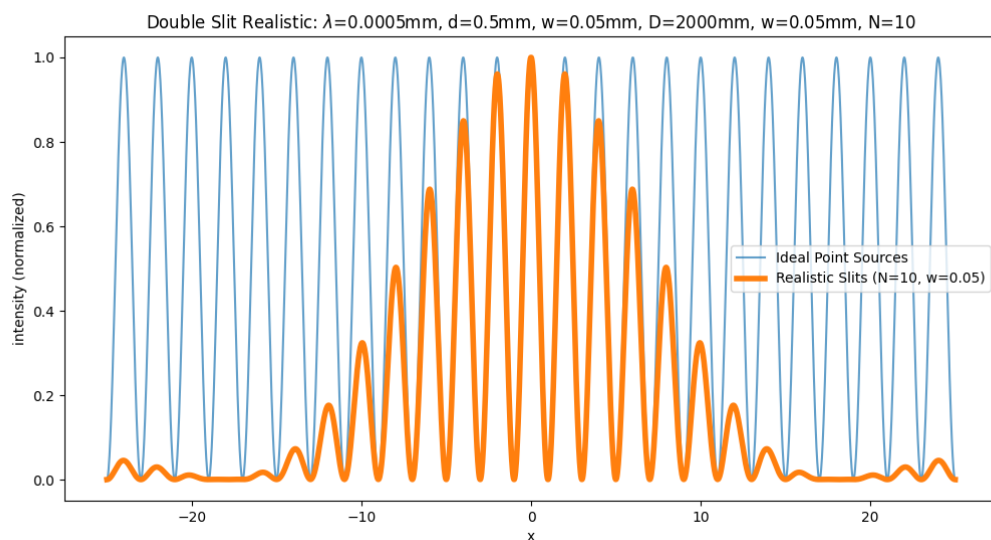
Results: As seen in Figure 3 ($w = 0.05$ mm), we can observe that a greater N implies a higher intensity peak. For example, for $N = 2$ the peak is much lower than for $N = 10$. We then varied w as shown in Figure 4. The figures indicate that a smaller w (0.05 mm) results in a wider peak where the central maxima dominates, while a greater w (0.2 mm) results in narrower peaks that are packed more tightly together.

Q4: Realistic Double Slit Model

Method: We copied 2 of the "realistic" slits from problem 3 ($N = 10$, $w = 0.05$ mm) and separated them by $d = 0.5$ mm and compared it to the point-source model.



(a) Point source model



(b) Realistic slit model

Figure 5: Comparison between point source double slit ($d = 0.5$ mm, $\lambda = 0.0005$ mm, $D = 2000$ mm). and realistic double slit (with discretization along a continuous width $N = 10$, $w = 0.05$ mm),

Results: What can be observed from the graphs in Figure 5 is that they seem to agree on the whole. Both the point source model and the realistic model have similar distances between their peaks. The main difference is that the realistic model's peaks get smaller further from the center.